The Bayesian Philosophy A New Way of Thinking Transcript

Hi everyone, let's learn about Bayesian inference. This is possibly your new way of thinking about probabilities and you will soon learn about why I am saying so. But think of Bayesian inference as inferring about uncertainties because the term inference has involved there right.

So inferring about uncertainties using some prior knowledge or the data that you have got. So that's the entire philosophy behind Bayesian approaches. You start with some belief, you update your belief as you see new data and then you proceed with that.

Let's begin and let's explore more about what exactly this entire concept is. So almost everything that you have learned till now is a part of frequentist approach or frequentist perspective. You have tests, certain types of tests, you have certain types of hypothesis, you lay down hypothesis, you perform test, you see p-values and then you will see whether the hypothesis that you have built is right or wrong and then you also hypothesise about your point estimates, build confidence interval right.

On the left hand side, if you see the frequentist view, you have the probability is the long run frequency of an event, a parameter like the mean okay is fixed, unknown constant right and data is random. Here in this approach we are saying the parameter like mean or standard deviation, they are all fixed, they are unknown constant and data is random. Now what we do with that data, the samples that we collect is we try building 95% confidence interval.

So this confidence interval we think of if we repeat the experiment n number of times there would be 95% of the experiments wherein the true value of the parameter would lie. That's what the view of confidence interval is but the Bayesian approach, in the Bayesian approach the probability is not seen as a long run frequency of an event, they are seen as the degree of belief or disbelief okay or you can say the confidence about a statement that you have okay. A parameter is random, here in the parameter is not fixed or it is unknown constant that is for sure because in both of the approaches we are dealing with something which is unknown but here the parameter is not fixed okay, the parameter is random and it is a random variable that has a specific type of probability distribution and then we try to find out 95% of probability that the parameter lies between certain range okay.

So that's the entire view of Bayesian, Bayesian we start with a random variable, we build a nice looking probability distribution or we assume about the probability distribution because we get some sense of the data, using that data we build probability distribution and we use that probability distribution to then find out the range within which the actual value of the parameter could lie. So that's the Bayesian way of doing that okay, we have prior belief, you start with some belief initially, so this is where you have not seen the data yet okay, you start with some belief let's say for example you are trying to detect a thief and you know who could be a thief given you know about the city okay, now that you have got a new evidence that the thief's foot size was somewhere around 6 feet, no not 6 feet but let's say 6 inches right, now given the new information that you have got, how likely is that you are going to catch the thief, so probability of catching a thief given you have the new information which is the size of the feet of the thief, now all of this information you can easily write it as P probability of finding a thief given you have a new evidence right, so you have got the new data which serves as a new evidence for you, now your entire belief would change because you have got a new data right, previously you might have had a probability that thief is from a particular area right, now given that you have got the shoe size of the thief, you know that most of the long shoes size people they are not living in particular region, so your entire belief would change based on what new information that you have got right, I am just telling you about an example you can think about any other example but think of it in this way, you start with the prior belief about something, as soon as you get new information or new data you then change your belief, this entire approach is called as Bayesian or you have already seen that in Bayesian theorem probability of A given B is equals to probability of B given A into probability of A divided by probability of B right, so in that case evidence is what you get in the denominator because that is what is changing your entire information, probability of A given B, B is the new information right, so think of it in this way, you have on the left hand side this is conditional probability but you can also write the same in Bayesian terms let's say here right, so I have not written it here but let me go to foundation section and let me be here, so let's say for example you have probability of B given A, now probability of B given A given that you have new information already or you have received the new information already right, let me help you understand this in a new way, let's say every single day you have a probability of getting spam emails as 80% which means that out of 100 emails that you are getting 80 are spam emails, now given that a new email has come on your mobile and also given that you have received an email from your boss, now majorly you will not receive a spam email from your boss right, so because of this new information that the email is from your boss your probability of spam may reduce to 10% or may reduce to even closer to 0 right, so this is the entire belief system that we built using Bayesian, so Bayesian is probability of having something or knowing about any uncertainties changing that probability given we have received a new data, new evidence, so this is entire Bayesian approach that you have already learned I don't know whether you have given it a thought in this way or not but you should think it from that perspective because every single thing that you learn in the formula for Bayesian is has been represented here, you start with a prior which is what you have on the numerator side of your right-hand side of the equation and then you have the data or the evidence which is at the denominator, you update your belief based on your evidence, so that's what it is okay, now this this entire Bayesian inference can be compared or the frequentist and the Bayesian approaches can be compared using confidence versus credibility okay, a frequentist approach calculates a 95% confidence interval for the mean weight of a product, this means that if we tend to repeat the sampling process many number of times, 95% of the calculated intervals would contain that true fixed value of the weight okay, this is what you have already learned and in this case we cannot say that there is 95% probability that true mean is in our specific interval and if you actually cannot do that but in Bayesian we instead of building a confidence interval we build a credible interval, so a Bayesian calculates 95% credible interval, this means given our data and our prior beliefs okay, there is 95% probability that the true value of the mean or the true value of the weight or mean of the weight lies within this interval, this interpretation itself is directly and intuitively related to some somewhat to confidence interval but this shouldn't be mistaken with what confidence interval is, in Bayesian we build a credible interval not a confidence interval and we use probability distributions to help us build that okay, because if you remember in the last section I told you that we start with a random variable, we build up a probability distribution around it and then we use that probability distribution to build 95% confidence interval or credible interval around it so that's what this entire Bayesian approach is, I hope this entire section was helpful for you to understand or to differentiate between Bayesian and Frequentist approach.

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